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# The Spillover Effect of Fraudulent Reviews on Product Recommendations

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**Abstract.** As the prevalence of user-generated reviews has been growing, the pervasiveness of fraudulent reviews has been increasing as well. In an effort to alleviate the consequences of fraudulent reviews, platforms have been using machine-learning algorithms for fraudulent review detection. However, the current business practice of simply removing fraudulent reviews might not be sufficient, as even their temporary presence might forge spillover effects propagating through other shopping tools. In particular, we examine and discover the persistence of long-lasting significant adverse impact of fraudulent reviews through their propagation to recommender systems, even long *after* successfully detecting and removing *all* fraud incidents. We conduct additional analyses further examining the intensity and evolution of the spillover effect over time across different dimensions, such as the cost of the fraudulent activity, the effectiveness and timeliness of the detection algorithms, and the quality of products. The results illustrate the gravity of the identified effect and highlight that the current business practice of just removing the fraudulent reviews is insufficient in the presence of recommender systems, albeit potentially sufficient in their absence in the long run. Such findings are timely as they might inform current regulatory and policy discussions. Finally, we also design a simple remedy that can be easily introduced to collaborative filtering algorithms to debias the recommender systems and demonstrate its effectiveness in alleviating the impact of the identified spillover effect.

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## 1. Introduction

The prevalence of online product reviews has been rapidly growing over the years, with consumers frequently consulting them when searching online for products. Thanks to this prevalence and the richness of the provided information, user-generated reviews now constitute an indispensable part of the online shopping experience (Todri et al. 2022), with more than 82% of consumers reading reviews before purchasing a product (Statista Research Department 2020) and trusting them as much as personalized suggestions (Murphy 2018). Online reviews, apart from being a valuable source of information for consumers, are also associated with an increase in consumer spending (Activate 2018).

The impact of product reviews on consumer behavior has, unfortunately, also fueled the emergence of fraudulent reviews, as indicated by the increasing sellers' efforts to commit review fraud and gain monetary advantages (Hern 2021). Recent research has documented the markets for fake reviews where dishonest product sellers can purchase fake reviews accruing substantial benefits due to the increased ratings in legitimate shopping platforms, such as Amazon.com (He et al. 2020). Although the pervasiveness of fake reviews is not yet fully understood, media and researchers alike report that a significant number of products and sellers are nowadays associated with fake reviews (Mayzlin et al. 2014, Luca and Zervas 2016, Sterling 2018, Hern 2021).

This surge of fraudulent reviews has instigated broad public concerns and, consequently, platforms have started cracking down on such activity. For instance, Amazon.com has banned paying for reviews while also using sting operations and machine-learning algorithms to detect fraudulent reviews (Luca and Zervas 2016, Sterling 2018). The current practices of online platforms pertain to removing any detected fraudulent reviews, without taking any further actions (Lappas et al. 2016, Ananthakrishnan et al. 2020).

However, although deleting the identified fraudulent reviews (Ananthakrishnan et al. 2020) might alleviate the direct benefits for fraudulent sellers (by correcting the inflated average product ratings), this might still not be completely sufficient, leading to adverse consequences for consumers and platforms as other shopping assistance mechanisms and tools might also be affected by the mere temporary presence of fraudulent reviews. The reason for our concern is that fraudulent reviews might affect consumer behavior, while platforms use data on this consumer behavior as *input* to other shopping assistance tools, such as recommender systems (RSs) (Adomavicius and Tuzhilin 2005, Adamopoulos 2013). Hence, spillover effects might be created, propagating lasting biases in favor of products associated with fraudulent activity. One could expect that such spillover effects *dissipate completely* after the successful detection and removal of fraudulent reviews and, hence, do not have a long-lasting impact. However, prior research has not examined whether fraudulent reviews continue to affect recommender systems and, consequently, users even after successfully detecting and completely eliminating all review fraud.

Recommender systems, in particular, are a valuable tool for consumers as, similar to user-generated product reviews, they help overcome information asymmetry and hence drive a substantial share of product sales (Gomez-Uribe and Hunt 2016, Dzyabura and Hauser 2019, Todri 2022). Prior research has indeed established that recommender systems are an essential part of the consumer journey and influence consumer behavior, leading to increased product sales (Senecal and Nantel 2004, De et al. 2010, Kumar and Hosanagar 2019). Literature has also shown that recommender systems can also create reinforcing loops leading to commonalities among user choices and demand concentration (Adamopoulos and Tuzhilin 2014b, Hosanagar et al. 2014, Adamopoulos et al. 2015). However, prior work has not examined whether fraudulent reviews lead to lasting biases in algorithmic recommendations even after successfully detecting and removing all fraudulent activity from the platform. Hence, we study whether such spillover effects might continue benefiting the corresponding products and sellers after the fraudulent reviews are removed. The impact of recommender systems in combination with the increasing prevalence of fake reviews and the upcoming interventions of policymakers (CMA 2021) render this a timely and vital research question in need of scientific investigation.

In particular, we examine whether such spillover effects of fraudulent reviews on product recommendations emerge, despite seemingly effective managerial efforts, using a simulation framework to fully control the fraudulent activity and establish necessary ground truth. Based on the conducted analyses, we discover that the adverse effects of review fraud on consumer purchases are further amplified via their propagation through recommender systems. Interestingly, we also discover that *the economically and statistically significant long-lasting adverse spillover effect of fraudulent reviews on recommender systems persists even long after successfully detecting and removing all fraud incidents*. We also conduct additional analyses examining the intensity of the spillover effect over time across different dimensions, such as the cost of the fraudulent activity, the effectiveness and timeliness of the detection algorithms, and the quality of products. The results illustrate the gravity of the identified effect and highlight that the current business practice of simply removing the fraudulent reviews is not sufficient in the presence of recommendations, albeit potentially sufficient in their absence in the long run. Hence, our work contributes to the literature on fake reviews and, in particular, the stream of work on *fraudulent review management* as it uncovers that, despite the current fraudulent review management efforts, an important spillover effect is left unaddressed.

In addition, we also *design and apply a simple but effective remedy* that can be easily introduced to current recommender systems to alleviate the impact of the spillover effect of fraudulent reviews debiasing the platform recommendations. This remedy weights the input to each algorithm relative to the probability that each transaction has been influenced by fraudulent reviews and is found to be effective for all examined algorithms. Hence, our work also contributes to the literature on recommender systems as the proposed remedy can be applied to state-of-the-art algorithms.

Our findings have managerial implications of interest as they illustrate a major shortcoming of current fraudulent review management practices and can guide future efforts toward alleviating the adverse effects of review fraud and debiasing platform recommendations. Finally, this study is also timely as policymakers currently start investigating fraudulent reviews considering their interventions (FTC 2019, CMA 2021, DSA 2022), and thus our findings can also inform imminent regulatory and policy discussions.

## 2. Related Work

Prior work has extensively examined the influence of user-generated content and online reviews on consumer behavior (Adamopoulos and Tuzhilin 2015, Adamopoulos et al. 2018). The importance and prevalence of user-generated reviews has recently given rise to fraudulent reviews that aim to deceive consumers into purchasing the corresponding products.

### 2.1. Fraudulent Reviews

In the emerging stream of literature on fraudulent reviews, the seminal work of Mayzlin et al. (2014) illustrates that consumers might indeed be fooled by fraudulent reviews and make suboptimal choices subsequently. Extending this line of research, Luca and Zervas (2016) demonstrate that review fraud is driven by such seller incentives, and He et al. (2020) document extensively the existence of large online markets for fake reviews. Focusing on fraudulent review management policies, Lappas et al. (2016) and Lam and Riedl (2004) examine the effectiveness of different fraud strategies, whereas Ananthakrishnan et al. (2020) find that a policy of flagging fake reviews but leaving them posted, instead of censoring them, may increase consumer trust in the platform. Regarding fraudulent review management, there is also related work on computer science on how to detect fraudulent reviews and shilling attacks (Burke et al. 2006, Jindal et al. 2010, Hu et al. 2012, Lappas 2012, Zhou et al. 2018). For instance, Mukherjee et al. (2012) achieve an accuracy of 86% in detecting fraudulent reviews, whereas nowadays, platforms could potentially use more accurate detection algorithms (Ananthakrishnan et al. 2020) or methods for faster detection (Zhong et al. 2021). However, prior work has not examined whether, despite the usage of fraudulent review detection and management techniques, there are any spillover effects of fraudulent reviews on other shopping tools, such as recommender systems.

In this study, we draw on the above literature and examine the spillover effect of fraudulent reviews on recommender systems, rather than the direct effect of such reviews.

### 2.2. Recommender Systems

Our work is also related to the recommender systems literature (please see Ricci et al. (2015) or Li and Karahanna (2015) for a complete literature review). Recommender systems have achieved broad social and business acceptance over the past decades and have been extensively studied in the computer science and business fields. The most prevalent approaches in both the industry and the literature follow the collaborative filtering paradigm (Adomavicius and Tuzhilin 2005). Collaborative filtering approaches have their roots in stereotyping (Rich 1979) and are based on the principle of using other users' information to predict the preferences of the focal user. Typical examples of such algorithms are the nearest neighbors (Koren 2010) and matrix factorization (Mnih and Salakhutdinov 2007) methods, which are successfully used in companies such as Amazon (Linden et al. 2003) and Netflix (Gomez-Uribe and Hunt 2016).

Beyond technical aspects such as predictive accuracy and concentration bias (Fleder and Hosanagar 2009, Adamopoulos and Tuzhilin 2013, Ricci et al. 2015), prior research has also examined the effect of RSs on consumer behavior and product demand (Oestreicher-Singer and Sundararajan 2012, Jabr and Zheng 2014, Kumar and Hosanagar 2019, Li et al. 2021, Adamopoulos et al. 2022, Mousavi et al. 2023). For instance, De et al. (2010) and Dias et al. (2008) find that RSs are associated with an increase in aggregate sales. Moreover, Brynjolfsson et al. (2011) provide empirical evidence that RSs are associated with increased aggregate sales diversity, reflecting lower search costs, and Hosanagar et al. (2014) find they widen users' interests, which in turn creates commonality with other users due to concentration bias. However, prior research has not examined whether fraudulent reviews affect recommender systems and, consequently, users even after successfully detecting and removing fraudulent reviews.

Our study contributes to the above literature by uncovering the existence of a spillover effect of fraudulent reviews on recommender systems and its long persistence, even after successfully removing all such fraud. This study also makes a secondary contribution as it proposes a remedy for debiasing recommendations and alleviating the impact of the spillover effect of fraudulent reviews.

## 3. Market-Based Simulation Design

We examine the possibility of a spillover effect of fraudulent reviews on RSs and the corresponding effectiveness of detecting and deleting such reviews by simulating an online platform using current real-world market conditions (Sections 3.1–3.3). Conducting extensive simulations entails important benefits for this study. First, it allows us to fully control the fraudulent activity establishing necessary ground truth. Second, it allows us to examine the time-evolving and nonlinear nature of the effect of interest under both current market conditions and different what-if scenarios and responses from online platforms. Besides, simulations will enable us to overcome a typical limitation of literature on fake reviews while preserving the privacy of the users; the absence of observational data are a common limitation because of the sensitive nature of economic transactions and personalized recommendations, and the lack of ground truth for fraudulent reviews (Mayzlin et al. 2014, Lappas et al. 2016, Ananthakrishnan et al. 2020). In addition, combining simulations with consumer choice models and actual recommender systems allows us to realistically assess the proposed algorithmic remedy under various conditions. Similar simulation designs have been used extensively in the literature (Fleder and Hosanagar 2009, Maroulis et al. 2010, Prawesh and Padmanabhan 2014, Ricci et al. 2015).

### 3.1. Simulation Process

An overview of the process is as follows: Algorithm 1 provides the pseudocode for the main simulation process. There are in total $I$ items and $U$ users in the market. $U^f$ of the users are fraudulent reviewers, and the rest are typical (nonfraudulent) consumers. Similarly, $I^f$ of the items are associated with fraudulent reviewers, and the rest of the items are nonfraudulent (i.e., not related to fraudulent reviews).[1] Each item's quality is randomly determined at the beginning of the simulation; in additional analyses, we allow for the possibility that the fraudulent items are of lower quality than the rest of the items.

Every period, the platform generates recommendations based on past purchase patterns, as described in Section 3.2, and each consumer purchases a product. For nonfraudulent consumers, we model their purchases using a multinomial choice model, as described in Section 3.3. For fraudulent users, they select one of the products associated with fraudulent reviews and might purchase it with probability $p^{pr}$; in the main analysis, we assume fraudulent reviewers do not purchase any product in order to more clearly illustrate the spillover effect, while in additional analyses we allow for this possibility despite the associated purchase cost, following prior research (He et al. 2020).

Then, each nonfraudulent user might review the purchased product with probability $r^{pr}$ and assign a truthful rating (i.e., rounding the score to the nearest half-star rating of the actual, experienced, item quality), whereas each fraudulent reviewer assigns a five-star rating to the corresponding product selected in the previous step to maximize the seller's benefits (Hern 2021); prior literature reports it is unlikely to observe fake negative reviews to competitor products (He et al. 2020) as it is less efficient (Lappas et al. 2016).

After a predetermined period, $t^r$, the platform detects fraudulent reviews and removes them from the platform (He et al. 2020). Each fraudulent review is successfully detected and removed with probability $d^{pr}$; the precision of the platform is assumed to be 100% to illustrate the effect of interest more clearly, while in the additional analyses we examine how the spillover effect varies across levels of veracity. After the platform detects and removes the fraudulent reviews, there is no additional fraud activity (i.e., fraudulent users neither review nor purchase items) on the platform.

Finally, after a number of iterations, the procedure concludes and the distribution of purchases from consumers (excluding any purchases from fraudulent users) *after* the removal of fraudulent reviews is computed and compared across fraudulent and nonfraudulent items for different recommendation strategies.

In the conducted analyses, we set the total number of items to 100, of which 20 are associated with fraudulent activity; prior research indicates that about 23% of products are associated with fraudulent reviews on platforms such as Amazon.com (He et al. 2020), while the results are qualitatively the same across any parameter values. Similarly, there are 100 consumers and 20 fraudulent reviewers on the platform. We set the number of iterations to 500, the total number of (time) periods to 200, and the period of removing the fraudulent activity to 100; prior research has found that when fraudulent reviews are removed, there is a lag of over 100 days (He et al. 2020). In the main analysis, we also set the probability of a fraudulent review to be successfully detected to 100%, the number of product recommendations to 10, the probability of a nonfraudulent consumer to review an item to 2% (Quora 2012, Mellinas 2019), and do not allow fraudulent users to purchase any items, and assume the quality of items (and review ratings) range from zero to five. In the additional analyses, we further study the effect of interest when varying any parameter values, including the number of periods, time of removing the fraudulent reviews, probability of removing fraudulent reviews, quality of fraudulent items, probability of fraudulent users purchasing products, and so on. In the following paragraphs, we discuss in detail all additional components and parameters of the simulation.

**Algorithm 1** (Simulation Procedure)

**Input:** Recommender system algorithm $RS$, choice model $M$, etc.
**Output:** Distribution of recommendations

$q_i$: item $i$ quality
$\overline{q}$: maximum item quality
$k$: number of recommendations
$r^{pr}$: probability of a nonfraudulent user $u$ to review item $i$ after purchase
$p^{pr}$: probability of a fraudulent user $u$ to purchase item $i$
$d^{pr}$: probability of a fraudulent review to be removed from the platform
$R^f$: set of fraudulent reviews on the platform
$t^r$: time period platform removes fraudulent reviews
$U^f$: subset of fraudulent users, $U^f \subset U$
$I^f$: subset of items associated with fraudulent users, $I^f \subset I$

```
foreach iteration e do
    Draw items I, each of quality q_i ∈ [0,5];
    foreach (time) period t do
        foreach user u do
            Recommend to user u top k items based on recommender algorithm RS, if t > 0;
            if u ∉ U^f then
                User u purchases item i based on multinomial choice model M;
                Draw r_{u,t} ∈ [0,1];
                if r_{u,t} < r^pr then
                    User u truthfully reviews item i with rating round(q_i/.5) * .5;
                end
            else if u ∈ U^f and t ≤ t^r then
                Draw i from I^f and p_{u,t} ∈ [0,1];
                if p_{u,t} < p^pr then
                    User u purchases item i;
                end
                User u reviews item i with rating q̄;
        end
        if t == t^r then
            foreach review r_{u,i} ∈ R^f do
                Draw d_{i,r} ∈ [0,1];
                if d_{i,r} < d^pr then
                    Delete review r of user u for product i;
                end
            end
        end
    end
end
```

### 3.2. Recommender System Algorithms

The current study focuses on collaborating filtering recommender systems, as this is the most prevalent approach in both industry and academia (Linden et al. 2003, Adomavicius and Tuzhilin 2005, Gomez-Uribe and Hunt 2016). In particular, we examine recommendations based on the two most popular methods: namely, *k*-nearest-neighbors (*k*NN), which is used on platforms such as Amazon (Linden et al. 2003), and matrix factorization (MF), which is used on platforms such as Netflix (Gomez-Uribe and Hunt 2016). It is worth noting that these methods not only represent different algorithmic families and are the most widely used approaches in practice (Adomavicius and Tuzhilin 2005), but they also exhibit different levels of susceptibility to different consumption biases in the absence of any interventions or fraudulent review detection and management efforts (Adamopoulos and Tuzhilin 2014a, Adamopoulos et al. 2015). We also include nonpersonalized recommendations based on best-sellers; the best-sellers recommendations are determined based on the (descending) ranking of products in terms of past sales. For a richer comparison, we also include specifications without any recommendations to users as well as specifications where the recommended items are randomly determined (Li et al. 2021). In the conducted analyses, we use the top 10 recommendations, five neighbors for *k*NN, and 10 latent factors for MF following the related work (Ricci et al. 2015), and all algorithms are trained at the end of each time period; the results remain qualitatively the same for different numbers of neighbors and latent factors.

We also propose in Section 5 a simple algorithmic remedy that platforms can apply to collaborative filtering algorithms to alleviate the adverse effect we identify and examine.

### 3.3. Product Choice Model

At each (time) period, users might purchase one of the available items; the results remain the same including an "outside" or no-purchase option. We model this using a multinomial logit choice model (Train 2009). The logit choice model is well established in economics and marketing research with strong theoretical foundations and models consumer choices based on two components, a deterministic component that includes the relevant product attributes and a random component that is independent and identically distributed with extreme value distribution (Train 2009); the results remain the same using a probit model. The deterministic component in our study includes the product "quality" (vertical differentiation) for the corresponding product (from zero to five) and whether the product is recommended to the corresponding user (either zero or one), with weights of 0.3 and 0.8, respectively; these estimates are based on prior research jointly examining recommendations and reviews (Adamopoulos et al. 2022).[1] For the vertical differentiation (product quality), the truthful quality ($q_i$) of a product is used when the corresponding consumer has purchased (experienced) the product in any previous period, whereas the average product review rating for the corresponding product is used as a proxy when a consumer has not previously purchased the product. The deterministic component can be further extended to include additional attributes, such as price and marketing promotions; in our simulation process, such attributes are encompassed in the "quality" score following prior research (Fleder and Hosanagar 2009), and the results remain qualitatively the same when including these as separate attributes, when including other additional attributes such as the number of reviews and sales, or when user preferences are heterogeneous and items are horizontally differentiated, as discussed in Section 4.2.6 and Online Appendix B. Finally, following the extant literature (Fleder and Hosanagar 2009), repeat purchases are not prohibited; examples of settings with repeat purchases include, for instance, music and video streaming as well as retail stores.

### 3.4. Measures of Interest

Finally, to examine the effect of interest, we introduce a measure that captures each product's average sales after the (time) period of detecting and removing the fraudulent activity; note that there is no additional fraud after this period. In particular, we measure average product sales for products associated with fraudulent reviews after the removal of fraudulent reviews as $\frac{\sum_U \sum_{I^f} \sum_e^E \sum_{t>t^r}^T \text{purchase}_{u,i,e,t}}{|U||I^f|E(T-t^r)}$, where $E$ is the total number of iterations; we similarly measure the average sales of the remaining products to facilitate the comparison. We also develop similar measures for the number of recommendations and further verify the underlying mechanism.

## 4. Results

We now present the empirical results examining the existence of a spillover effect of fraudulent reviews on recommendations, even *after* the successful detection and removal of *all* fraud incidents.

### 4.1. Main Results

As Table 1 shows, items that were associated with fake reviews in the past ($t \leq t^r$) have on average significantly higher sales after the successful detection and removal of all fake reviews ($t > t^r$); Figure 1 visualizes the corresponding boxplots to better illustrate the differences. A $t$ test of paired differences for unequal means shows that the differences are significant. That is, items that were associated with fraudulent reviews

**Table 1.** Comparison of Sales of Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Recommender System Algorithms After Removing All Fake Reviews from the Platforms

| Recommendations | Items with fake reviews | Items without fake reviews |
|---|---|---|
| No recommendations | 0.01019 | 0.009953 |
| | (0.00085) | (0.00021) |
| Random recommendations | 0.010249 | 0.009938 |
| | (0.00085) | (0.00021) |
| *k*NN recommendations | 0.012449 | 0.009388 |
| | (0.00152) | (0.00038) |
| MF recommendations | 0.014844 | 0.008789 |
| | (0.00137) | (0.00034) |
| Best-seller recommendations | 0.014688 | 0.008828 |
| | (0.00155) | (0.00039) |

*Notes.* Five hundred iterations with $t^r = 100$, $d^{pr} = 1$, $p^{pr} = 0$, and $k = 10$. See Section 3.4 for the exact measures.

in the past continue to have higher sales even after the removal of these reviews due to the bias introduced before their removal. In particular, the sales of items associated with fraudulent reviews, compared with the rest of the items, are about 32.6%, 68.9%, and 66.4% higher in the cases of kNN, MF, and best-seller recommendations, respectively.[2] These figures also illustrate the economic significance of the identified spillover effect. This is a very interesting finding, as in our main analysis we assume that the platform has a success rate ($d^{pr}$) of 100% in detecting and removing fake reviews and the fraudulent users do not purchase any of the items, but the adverse effects of fraudulent reviews still propagate to recommender systems.

It is also worth noting that there is a difference in the cases of no recommendations and random recommendations.[3] This (statistically nonsignificant at $t = T$) difference is because items associated with fraudulent reviews become more easily discoverable by users as they attract nonfraudulent reviews (from legitimate consumers) due to increased sales; the results are qualitatively the same when assuming the average rating in the marketplace is used as a proxy when there are no ratings for a product or that all products have available ratings before the introduction of fraudulent reviews. This observation that the spillover effect is amplified in the cases of *k*NN, MF, and best-seller recommendations compared with the cases of no recommendations and random recommendations confirms the spillover effect is due to the recommender systems algorithms and demonstrates that the business practice of removing the fraudulent reviews is not effective when the platform uses recommendation algorithms while providing evidence regarding the underlying mechanism. That is, the previous statements further illustrate that the identified spillover effect is due to recommender systems, and *the current business practice of merely removing the fraudulent reviews is not sufficient in the presence of recommendations*, albeit potentially sufficient in their absence in the long run under certain conditions. It is also interesting to note that we find that MF has a larger bias than *k*NN and comparable to best-sellers recommendations. This is likely due to the global nature of the MF approach, in contrast to the local nature of the *k*NN approach (Adamopoulos and Tuzhilin 2014b, Ricci et al. 2015); we further discuss this observation in Section 4.2.2.

We next examine the effect of interest across different time horizons ($T$) to better understand how fast it dissipates across different algorithms. Figure 2 shows the difference in sales between the items associated with fraudulent reviews and the items not related to such reviews (i.e., difference of measures in Section 3.4 for $I^f$ and $I \setminus I^f$) across different (time) periods. Interestingly,

**Figure 1.** (Color online) Comparison of Sales of Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Recommender System Algorithms After Removing All Fake Reviews

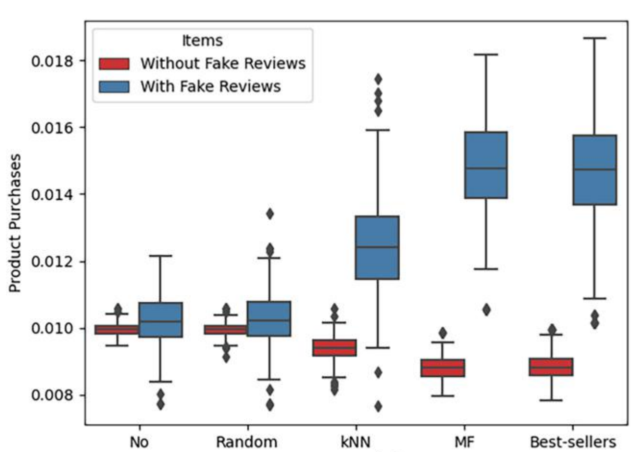
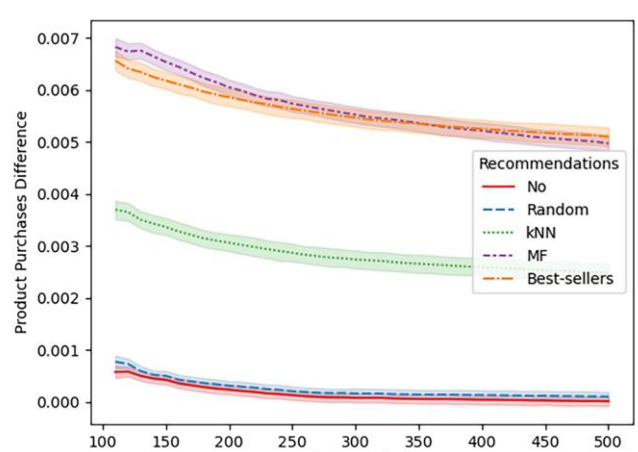



**Figure 2.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Time Horizons and Recommender System Algorithms After Removing All Fake Reviews



*the spillover effect of fake reviews on recommender systems survives over long horizons*, despite removing all fraudulent reviews from the platform and retraining all algorithms each period. In particular, even after 400 periods (e.g., days) after the detection and successful removal of all the fraudulent reviews from the platform, the sales of items associated with fraudulent reviews, compared with the rest of the items, are about 26.2%, 55.4%, and 57.0% higher in the cases of *k*NN, MF, and best-seller recommendations, respectively; the corresponding differences are about 0.0025, 0.0050, and 0.0051 in the instances of *k*NN, MF, and best-seller recommendations (the expected average product purchases are 0.01 given the corresponding measure in Section 3.4 and the number of available choices). This finding further illustrates the necessity of additional actions by the platforms when detecting the fraudulent activity, as discussed in Section 5, because the identified spillover effect does not dissipate over time. It is interesting to also note that the spillover effect of fake reviews on recommender systems persists over long horizons, even though the recommendation algorithms are retrained every (time) period (Linden et al. 2003); the spillover effect is expected to be even larger when the algorithms are retrained less frequently. Figure A1 in the online appendix illustrates the spillover effect for an even longer time horizon (i.e., 5,000 periods), further corroborating the previous discussion as the effect of fake reviews on recommender systems still does not fade away after the removal of the fake reviews, whereas the corresponding difference dissipates over time in the cases of no recommendations and random recommendations.

Finally, Figure A2 in the online appendix shows the number of times each item was recommended on average for items with and without fake reviews using the corresponding measures discussed in Section 3.4. As Figure A2 shows, items associated with fraudulent reviews are recommended more frequently in the cases of *k*NN, MF, and best-seller recommendations (with the increase being again larger for MF and best-sellers), whereas there is no difference in the cases of no recommendations and random recommendations. Hence, the results further confirm the underlying mechanism as the increased sales of items associated with fraudulent reviews are due to the more frequent recommendations, even after the successful detection and removal of all fraudulent activity.

## 4.2. Additional Results

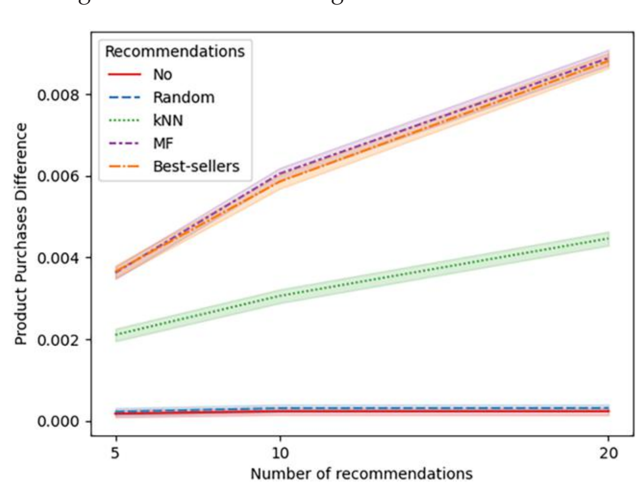
To further illustrate the strength of the identified spillover effect and the underlying mechanism, we examine the effect across different dimensions that might be determined by the platform or fraudulent users.

**4.2.1. Length of Recommendation Lists.** We first examine the spillover effect of interest across various lengths of recommendation lists ($k$). In the main analysis, we set the number of recommendations to 10, as this is a common choice for platforms. However, platforms might decide to deliver to users more or fewer recommendations. Hence, we next examine the effect of interest when generating instead 5 or 20 recommendations for each user in each period. Because the identified effect propagates via the platform's recommendations, one would expect that, according to the discussed mechanism, the longer the recommendation lists, the stronger the spillover effect, as there are more recommendation opportunities for items associated with fraudulent reviews.

As Figure 3 (and Table A.1 in the online appendix) shows, the results confirm the previous hypothesis and corroborate the main findings and the underlying mechanism. In particular, the *difference in sales between the items associated with fraudulent reviews and items not related to such reviews is indeed increasing with the length of recommendation lists*. As shown in Table A.1, when the length of recommendation lists increases to 20, the sales of items associated with fraudulent reviews, compared with the rest of the items, are about 49.1%, 108.1%, and 107.0% higher in the cases of *k*NN, MF, and best-seller recommendations, respectively. Beyond confirming that the spillover effect is indeed due to the recommender systems, this finding also illustrates simply reducing the length of recommendation lists does not alleviate the corresponding biases and potentially necessitates an algorithmic remedy.

**4.2.2. Cost of Fraudulent Activity.** Moreover, in the main analysis, fraudulent users do not purchase any of the associated items they review in order to minimize their cost. We repeat the analysis varying the probability of a fraudulent user to purchase an associated item

**Figure 3.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Lengths of Recommendation Lists and Algorithms After Removing All Fake Reviews



he/she reviews (i.e., $p^{pr}$ in Algorithm 1) from 0% to 100%.

As Figure 4 (and Table A.2 in the online appendix) shows, *when fraudulent reviewers purchase the corresponding items*, in addition to reviewing them, there is (a slight) additional increase in sales of these items (to nonfraudulent consumers) in the case of MF and best-sellers recommendations and *a significant additional increase for kNN recommendations*. In particular, when fraudulent reviewers also purchase the items they review ($p^{pr} = 1$), the sales of these items (to nonfraudulent consumers) after all fraud is stopped and the corresponding reviews are removed are about 47.4%, 72.2%, and 79.2% higher in the cases of *k*NN, MF, and best-seller recommendations, respectively. That is, for neighborhood (local) models the spillover effect mainly propagates via recommendations based on the purchase patterns of nonfraudulent consumers and further amplifies by some fraudulent users becoming nearest neighbors to nonfraudulent users based on their purchases, whereas for global models the purchases of nonfraudulent consumers are sufficient for the spillover effect to propagate via the recommendation algorithms. This observation is also illustrated in Figure A3 in the online appendix showing the number of recommendations for items with and without fake reviews across the different levels of $p^{pr}$ and confirming the aforementioned significant increase for local models, reducing the differences from global models; in local models, in contrast to global ones, recommendations are only affected by nearest neighbors.[4] Hence, the previous results further confirm the underlying mechanism and additionally illustrate that the increased sales of items associated with fraudulent reviews, even after the successful detection and removal of all fraudulent activity, is due to the more frequent recommendations mainly based on legitimate consumers' purchases. It is also interesting to note that this finding demonstrates that fraudulent sellers can further amplify the spillover effect of fraudulent reviews on recommender systems if they incur the cost of purchases; fraudulent users purchasing items they review may reduce the veracity of fraud detection.

**4.2.3. Veracity of Fraudulent Review Detection.** In addition, in the main results of this study, we set the platform's success rate in detecting and removing fraudulent reviews to 100% to illustrate the effect of interest more clearly. However, despite the advancements in the corresponding techniques (Mukherjee et al. 2012), platforms might achieve lower recall (i.e., fraction of successfully removed fraudulent reviews). Hence, we repeat the analysis varying the corresponding parameter, $d^{pr}$, to measure the spillover effect of fraudulent reviews on recommender systems more precisely. As Figure 5 (and Table A.3 in the online appendix) shows, the results confirm our main analysis and provide additional evidence further illustrating that the spillover effect propagates via nonfraudulent consumers' purchases. In particular, when the recall rate decreases to 85% ($d^{pr} = 0.85$), which is the lower bound for such state-of-the-art algorithms nowadays (Mukherjee et al. 2012), the sales of the items associated with fraudulent reviews further increase to about 37.8%, 74.4%, and 71.0% higher than the sales of the rest of the items in the cases of *k*NN, MF, and best-seller recommendations, respectively; *improvements in the recall rate or even perfect detection (with a lag) do not sufficiently alleviate the spillover effect*, further suggesting it propagates via nonfraudulent purchases. Thus, although detecting and removing

**Figure 4.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Levels of Probability of a Fraudulent User to Purchase an Item and Recommender System Algorithms After Removing All Fake Reviews

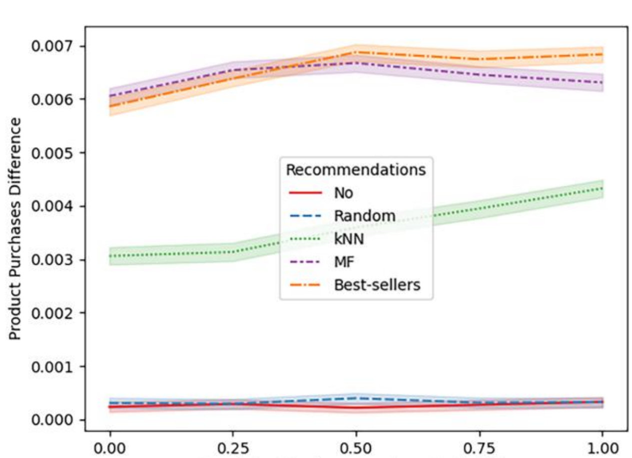
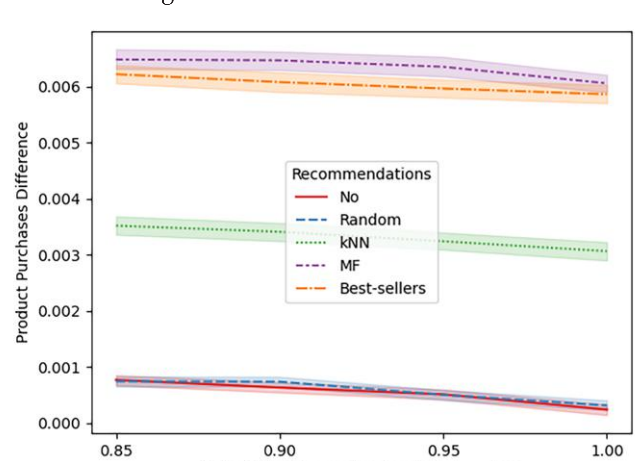



**Figure 5.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Success Rates (Recall) of Removing Fraudulent Reviews and Recommender System Algorithms After Removing All Fake Reviews

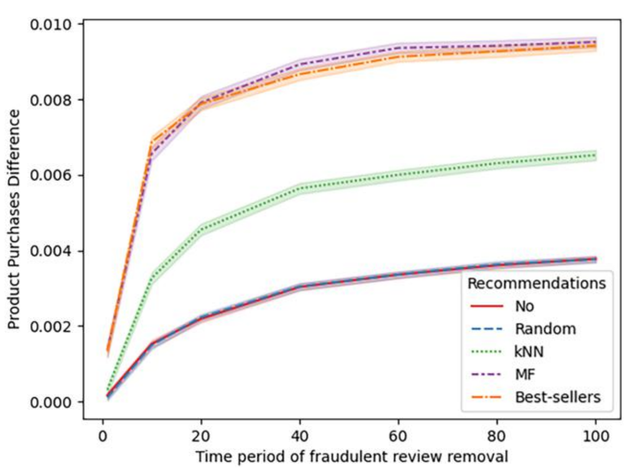


fraudulent reviews is not a sufficient remedy by itself, because the spillover effect propagates via nonfraudulent consumers' purchases, this finding illustrates the usefulness of using state-of-the-art algorithms to detect fraudulent reviews and shilling attacks as the spillover effect would have been even larger in the absence of such techniques.

In addition to examining this spillover effect across recall rates, we also repeat the analysis varying the precision as platforms might mistakenly detect nonfraudulent reviews as fraudulent. As Figure B1 in the online appendix shows, the robustness check results further confirm our main analysis.

**4.2.4. Velocity of Fraudulent Review Detection.** Furthermore, another dimension a platform might be able to affect is how fast the fraudulent reviews are detected and removed from the platform. In the main analysis, we set $t^r$ to 100, as the literature indicates that when fraudulent reviews are removed, there is a lag of over 100 days (He et al. 2020). As a robustness check, we vary this lag allowing the platform to detect fraudulent reviews earlier. As Figure 6 (and Table A.4 in the online appendix) shows, the results further corroborate our findings. In particular, when the detection lag reduces to just 20 periods from the typical of more than 100 days ($t^r = 20$), the sales of these items are still about 49.8%, 93.4%, and 92.9% higher until period $T$ in the cases of $k$NN, MF, and best-seller recommendations, respectively; we acknowledge that platforms might not be able to detect fraudulent reviews so fast as they often rely on time-consuming sting operations (Luca and Zervas 2016).[5] This finding further confirms the gravity of the identified spillover.

**Figure 6.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different (Time) Period of Initial Detection and Removal of Fake Reviews ($t^r$) and Recommender System Algorithms




Figures B2, B3, and B4 in the online appendix show the presence of the identified effect for alternative time windows.[6] The results also illustrate that the magnitude of spillover effect is increasing with $t_r$ (Figure B3) and that earlier (fraudulent) reviews have larger impact (Figure B4). Taking into consideration the previous findings, the robustness checks further illustrate that, *even if a platform responds with a reduced lag by employing state-of-the-art techniques, simply detecting and removing the fraudulent reviews is not by itself a sufficient management practice if there is any delay*; albeit faster responses reduce the magnitude of the adverse long-lasting effects of fraudulent reviews. Next, we examine the scenario of preposting review filtering (i.e., blocking), where the platform detects and removes the fraudulent reviews in real time before shown to any consumers.

***4.2.4.1. Immediate Preposting Fraudulent Review Blocking.*** Specifically, we also examine the identified spillover effect across different success rates of detecting and removing fraudulent reviews in real-time before being posted publicly on the platform (i.e., filtering before each review is shown to other consumers on the platform, see Footnote 6). Figure B5 in the online appendix shows the corresponding results for different levels of $F$ measure performance (precision = recall) of real-time detection and removal of fraud. For instance, even when the $F$ measure is 95%, the sales of items associated with fraudulent reviews, compared with the rest of the items, are about 29.8%, 49.7%, and 51.8% higher in the cases of $k$NN, MF, and best-seller recommendations, respectively. Hence, the results suggest that with imperfect detection simply removing (blocking) the fraudulent reviews is not a sufficient management practice, *even if a platform responds immediately* with preposting blocking. The results are qualitatively the same when the precision rate of real-time detection is 100% and we vary the recall rate.

It is also worth comparing immediate blocking with imperfect algorithms to the current practice of detection and removal with a lag. Hence, we also compare the magnitude of the identified spillover effect in each case across recall rates, for a state-of-the-art precision level of 85% for each case (Mukherjee et al. 2012, Zhong et al. 2021). Figure B6 shows the corresponding results suggesting that for a precision level of 85% a recall rate close to 99% would be needed for immediate detection to be more effective, whereas immediate preposting fraudulent review blocking might amplify the identified spillover effect for lower accuracy levels as nonfraudulent reviews are also blocked immediately due to the imperfect nature of detection algorithms limiting the ability of the system to self-correct to some extent based on early nonfraudulent reviews. These results also justify the current business practice of removing the detected reviews with a lag. In particular, when the detection performance is 85% precision and 90% recall, with preposting detection and blocking the sales of the items not associated with fraudulent reviews are reduced by 8.5%, 9.4%, and 8.6% compared with detection and removal of fraudulent reviews with a lag (and the same detection performance) in the cases of *k*NN, MF, and best-seller recommendations, respectively, due to the aforementioned reasons. Moreover, we also examine the identified spillover effect for a higher performance level (precision = 95%); we acknowledge that, in the case of immediate preposting filtering, platforms might not be able to achieve the state-of-the-art levels of precision and recall (recorded in applications with delayed detection), as a limited set of features may be available compared with the case of detecting and removing fraudulent reviews with a lag (e.g., missing subsequent reviews) (Mukherjee et al. 2012, Luca and Zervas 2016, Mewada and Dewang 2021, Zhong et al. 2021). Figure 7 (and Table A.5) shows the corresponding results illustrating that immediate detection is more effective (compared with removing detected reviews with a lag and the same detection performance) when recall is higher than 90% and as effective as detection with a lag for lower recall rates, for a precision rate of 95%. In particular, when the detection performance increases to 95% precision and recall (for both cases), with preposting detection and blocking the sales of the items associated with fraudulent reviews are reduced by 5.0%, 7.8%, and 9.6% compared with detection and removal of fraudulent reviews with a lag in the cases of *k*NN, MF, and best-seller recommendations, respectively. Thus, the results also suggest *platforms should surpass the current state-of-the-art performance in early detection* of fraudulent reviews in order for preposting blocking to be an effective remedy.

**4.2.5. Quality of Items Associated with Fraudulent Reviews.** Finally, we also allow the quality $q_i$ of the items that are related to fraudulent reviews ($I^f$) to be lower than the rest of the items. In particular, we vary the quality of fraudulent items from equal to half of the quality of the rest of the items; we operationalize this by multiplying the quality score determined in Algorithm 1 by $q^f \in [0.5, 0.6, \ldots, 1.0]$. As Figure 8 (and Table A.6 in the online appendix) shows, *due to the identified spillover effects, items associated with fraudulent reviews can achieve higher sales than the rest of the items, even when they are of significantly lower quality*. For instance, when the (average and maximum) quality of items associated with fraudulent reviews is 10% lower (i.e., $q^f = 0.9$), the sales of these items are still about 13.1%, 49.4%, and 45.1% higher than the sales of the rest of the items in the cases of *k*NN, MF, and best-seller recommendations, respectively.[7] This finding further illustrates the strength of the identified spillover effect and the significance of its implications for consumers and platforms.[8] It is also worth noting that, under current market conditions (i.e., items associated with fraudulent reviews constitute about 20% of the market and their quality is on average 5% lower; He et al. 2020), the identified spillover effect is remarkably strong as the sales of items associated with fraudulent reviews are 24.2%, 54.5%, and 55.3% higher than the sales of the rest of the items in the cases of *k*NN, MF, and best-seller recommendations, respectively.

**4.2.6. Additional Robustness Checks.** We further assess the robustness of the findings across all additional parameters of the estimation process. As discussed, and

**Figure 7.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Success Rates (Recall) of Removing Fraudulent Reviews with Immediate Preposting Fraudulent Review Blocking (i.e., `kNN, `MF, `Best-Sellers) or Delayed Detection and Removal (i.e., No, Random, kNN, MF, Best-Sellers) When Precision = 95% in Each Case

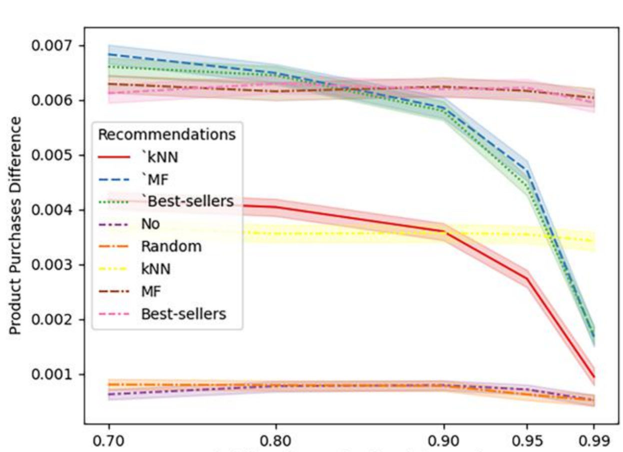
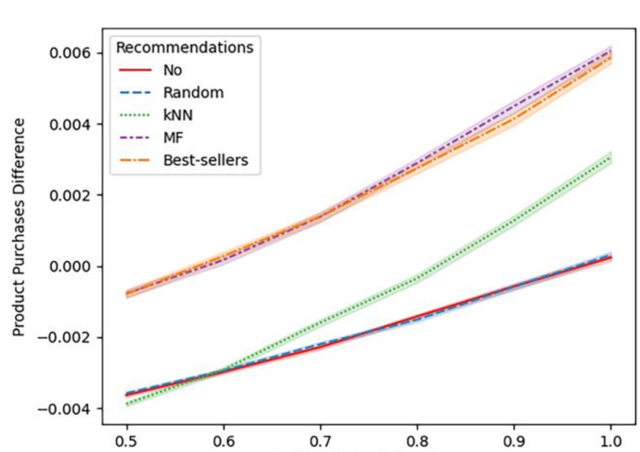



**Figure 8.** (Color online) Difference in Sales Between Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Quality Levels of Fraudulent Items and Recommender System Algorithms After Removing All Fake Reviews



shown in detail, in Online Appendix B, the results are qualitatively the same when incorporating additional item attributes (e.g., number of reviews and sales, separately from the vertical differentiation component) and heterogeneous user characteristics as well as different reviewing biases (e.g., under-reporting) or varying any other remaining estimation parameter, such as the number of items associated with fraudulent reviews and number of fraudulent reviewers. Overall, the robustness checks further illustrate that *simply detecting and removing the fraudulent reviews is not by itself a sufficient management practice in the presence of recommendations, regardless of the extent of the fraud and even when a platform responds quickly and employs state-of-the-art techniques*.

## 5. Recommender System Algorithms Remedy

After discovering the persistence of adverse long-lasting effects of fraudulent reviews, even after the perfectly successful detection and removal of all fraud incidents with a delay (or their immediate imperfect blocking at the time of generation), through their propagation to recommender systems, and following an investigation of several dimensions that the platform might be able to affect, we also conduct an additional analysis examining whether we can alleviate this bias by adapting the design of the algorithms. In particular, we design and apply a simple remedy that can be easily applied to collaborative filtering systems. Specifically, we postulate that, in addition to removing fraudulent activity, *nonfraudulent purchases that would not have occurred in the absence of fraudulent activity should be removed, too, from the recommendation algorithms*; otherwise, the spillover effect may also continue propagating due to transactions of nonfraudulent users that have been influenced by fraudulent activity, as discussed in Section 4.2. Exact attribution for such purchases requires ground truth and is characterized by tremendous challenges for marketplaces and impractical computational costs (Ghose and Todri-Adamopoulos 2016), necessitating the use of simple and efficient approaches. Hence, we propose to weight the input to each algorithm based on the probability that fraudulent reviews have influenced a transaction, using *observable to the platform factors*. Such remedy can be operationalized in different ways with factors *relating to the extent of fraudulent activity associated with each product* (e.g., number of detected fraudulent reviews, fraudulent increase in product rating). In what follows, we empirically test the proposed remedy by operationalizing this bias-reducing factor as deducting from each cell $r_{ui}$ of the transactions matrix $R$ the quantity $\min(r_{ui}, \text{floor}(c * r_{ui} * \hat{f}^{pr}))$, where $c$ is the penalty rate parameter, $\hat{f}^{pr}$ is operationalized using dimensionless quantities as $\left(\frac{\#\text{reviews}_i^f}{\#\text{reviews}_i} + \frac{\overline{\text{rating}_i^f} - \overline{\text{rating}_i}}{\overline{\text{rating}_i}}\right)$, $\#\text{reviews}_i^f$ is the number of detected fraudulent reviews for product $i$, $\#\text{reviews}_i$ is the total number of reviews for this product, $\overline{\text{rating}_i^f}$ is the average rating of the product in the presence of fraudulent reviews, and $\overline{\text{rating}_i}$ is the rating without the detected fraudulent reviews; the percentage of reviews that are detected as fraudulent and the corresponding increase in product rating are observable dimensionless factors that directly relate to the probability that fraudulent reviews have influenced nonfraudulent transactions (Cui et al. 2012, Xu et al. 2020). The previous bias-reducing factor is estimated on detecting and removing the fraudulent activity from the platform (i.e., period $t^r$) and is applied in every period afterward.

To empirically examine the effectiveness of the proposed remedy, we repeat the same analysis as before (Section 4.1), but we apply the proposed approach to all the used recommendation algorithms (i.e., *k*NN, MF, and best-sellers) as described previously; the penalty rate $c$ was determined by running the process on a smaller set with various parameter values and observing which value resulted in the desired outcome (i.e., alleviating the detected spillover effect) before conducting the simulations (Chae et al. 2019). As Table 2 shows, when incorporating the proposed remedy, items that were associated with fake reviews in the past now have on average the same sales as the rest of the items, after the successful detection and removal of all fake reviews; Figure 9 visualizes the corresponding boxplots to better illustrate the effectiveness of the algorithmic remedy. This finding demonstrates that platforms can alleviate the identified spillover effect by employing the proposed remedy. This is an interesting finding as the proposed remedy is more effective than the alternative examined strategies of earlier detection or imperfect prepost blocking; in contrast to alternative strategies, the proposed remedy considers nonfraudulent purchases that would not have occurred in the absence of

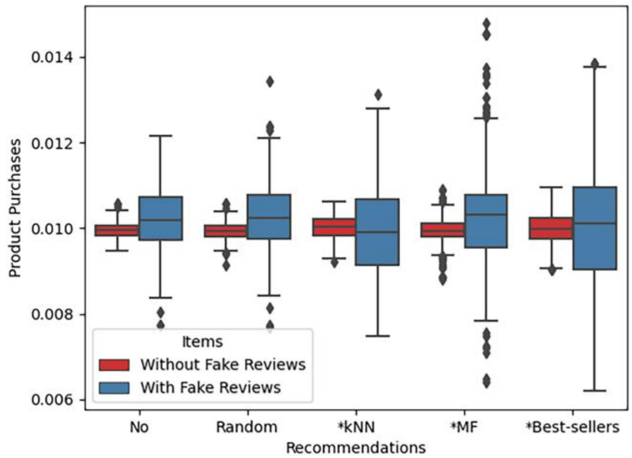
**Table 2.** Comparison of Sales of Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Recommender System Algorithms Incorporating the Proposed Remedy After Removing All Fake Reviews from the Platform

| Recommendations | Items with fake reviews | Items without fake reviews |
|---|---|---|
| No recommendations | 0.01019 | 0.009953 |
| | (0.00085) | (0.00021) |
| Random recommendations | 0.010249 | 0.009938 |
| | (0.00085) | (0.00021) |
| *k*NN recommendations[a] | 0.009921 | 0.010020 |
| | (0.00105) | (0.00026) |
| MF recommendations[a] | 0.010318 | 0.009921 |
| | (0.00117) | (0.00029) |
| Best-seller recommendations[a] | 0.010130 | 0.009968 |
| | (0.00155) | (0.00039) |

*Notes.* Five hundred iterations with $t^r = 100$, $d^{pr} = 1$, $p^{pr} = 0$, and $k = 10$. See Section 3.4 for the exact measures.

[a]The algorithm implements the correction discussed in Section 5.

**Figure 9.** (Color online) Comparison of Sales of Items Associated with Fake Reviews and Not Associated with Fake Reviews Across Different Recommender System Algorithms Incorporating the Proposed Remedy After Removing All Fake Reviews




fraudulent activity. Besides, the increase in the wall clock running time was less than 1%.

The results are qualitatively the same across settings and when varying any simulation parameter as discussed in Section 4.2. These settings include, among others, different choice models, varying veracity levels of fraudulent review detection, and differences in quality of items associated with fraudulent reviews. Figures C1–C10 in Online Appendix C present the corresponding results of the recommender systems remedy for the different parameter values and settings. The proposed correction effectively alleviates the bias from all examined algorithms, including *k*NN, MF, and best-sellers recommendations, regardless of the initial differences (Table 1; Figure 1). It is also especially noteworthy that the proposed approach alleviates the spillover effect of fraudulent reviews even in cases of imperfect detection. For instance, Figure C5 shows that, when incorporating the proposed remedy, items that were associated with fake reviews in the past now have on average the same sales as the rest of the items, even when the detection of fake reviews is imperfect and characterized by false positives as those are distributed across items; the proposed approach is also effective in the case of false negatives, by increasing the penalty rate compared with the case of perfect detection. Similarly, the proposed algorithmic remedy remains effective for different (time) periods of initial detection (Figure C6). Overall, as shown in Online Appendix C, the proposed remedy achieves similar performance across different parameter values and settings, while similar results were achieved using alternative transformations of the observable factors and varying the weights of the corresponding factors.

## 6. Discussion and Managerial Implications

As user-generated reviews are becoming increasingly prevalent in online shopping platforms, so do sellers' efforts to commit review fraud and gain monetary advantages. Thus far, there have been efforts to detect and manage such instances of fraud. Despite the considerable progress in these areas, we still lack academic and managerial understanding regarding the spillover effects of fraudulent reviews on other shopping assistance tools, such as recommender systems. Our work contributes to the literature on fake reviews as it studies the spillover effect of fraudulent reviews on recommender systems even after the successful deletion of such reviews. The findings yield timely implications extending our understanding of review fraud. For instance, our findings suggest that managers and researchers alike should further expand their fraudulent review management efforts to even longer time horizons, beyond the removal of fake reviews, as such review fraud might perpetually inadvertently affect consumer behavior even after the successful removal of all fraudulent reviews. Similarly, our analyses also suggest additional efforts for earlier and more accurate detection of fraud incidents. Importantly, the results also further deepen our understanding of the spillover effects of fake reviews and suggest that more holistic theories and approaches are needed considering shopping assistance tools vis-à-vis fraudulent reviews. For instance, the findings suggest fraudulent review management efforts should also expand to additional platform components, such as recommender systems. Such findings might also inform future regulatory and policy discussions. Policymakers might consider these spillover effects when determining fines and necessary platform interventions (FTC 2019, CMA 2021, DSA 2022). Our work also provides a framework that managers and stakeholders can use to estimate the spillover effect in the absence of ground truth; no party observes all fraudulent activities (Mayzlin et al. 2014, Lappas et al. 2016, Ananthakrishnan et al. 2020).

In addition, our work also contributes to the literature on recommender systems as we propose a simple remedy that can be applied to state-of-the-art algorithms and alleviates the corresponding bias in favor of products with fraudulent reviews. This finding also advances the state-of-the-art practice in the industry for managing fraudulent reviews. Platforms might use the suggested remedy to alleviate unaccounted biases of fake reviews on consumer behavior. Debiasing the recommendations provided to consumers by adopting such a remedy can also further deter any future fraudulent review activity by lessening sellers' incentives to commit fraud; toward this goal, the remedy may be used to further penalize any fraudulent review activity (see Online Appendix D). Future research may examine the extent to which such remedies and penalties deter fraudulent activity. Our analyses also suggest that future work may focus on proposing alternative remedies as well as enhancing the performance of early detection methods for fraudulent reviews. For instance, the findings regarding the potential remedy of preposting blocking may encourage future research to develop novel approaches using, in addition to the reviews, collaborative features and rich sources of information, such as product videos and images, based on deep-learning methods to advance the state of the art for early or immediate detection. Toward this goal, platforms may also enhance their efforts to establish ground truth utilizing crowdsourcing.

This work does not come without limitations. The primary limitation is the absence of observational data that includes ground truth for fraudulent reviews together with private economic transactions and personalized recommendations. This is a typical limitation for research on fake reviews, caused by the sensitive and proprietary nature of the underlying information of product purchases and personalized recommendations as well as the lack of ground truth for fraudulent reviews (Mayzlin et al. 2014, Lappas et al. 2016, Ananthakrishnan et al. 2020). Despite this limitation, this study advances the academic and managerial understanding of the long-term effect of fraudulent reviews in important ways and shows that this effect persists even after the successful detection of all review fraud. Nevertheless, this also suggests an avenue for future work. Finally, future work might also study potential spillover effects on other shopping assistance mechanisms and channels (Adamopoulos et al. 2020) and suggest relevant remedies to alleviate the corresponding biases.

## Endnotes

[1] We refer to the vertical differentiation measurement of each product as quality for illustrative reasons. We also explore alternative approaches in Section 4.2.6 and the results remain qualitatively the same.

[2] The difference in standard deviations between the estimates corresponding to items with fake reviews and items without fake reviews (for the same algorithm) is due to the different number of items in each group.

[3] The spillover effect may dissipate over a longer horizon in the cases of no recommendations and random recommendations when using alternative choice models as a robustness check, as discussed in Section 4.2.6 and Online Appendix B.

[4] We further confirm this observation by examining the spillover effect across different neighborhood sizes, as shown in Figure A4 in the online appendix; we acknowledge that the additional neighborhood sizes are larger than those in typical recommender systems applications (Ricci et al. 2015). The results illustrate that the difference between the local and global models is reduced as the neighborhood size of the local model increases; as the neighborhood size increases, the neighborhoods become broader and the local model tends to approximate a global one incorporating information from more users (Ricci et al. 2015).

[5] Setting $t^r$ to 0 with both precision and recall rates of 100% alleviates the adverse effects of fraudulent reviews, as this scenario is equivalent to the absence of any fraudulent activity.

[6] Figure 6 shows the difference in sales of items associated with fake reviews from time period 0 to $T$. Figure B2 in Online Appendix B shows the spillover effect from time period $t^r$ to $T$, with $t^r$ varying across conditions, whereas Figures B3 and B4 show the spillover effect from time period $t^r$ to $t^r + 100$ and from $t = 100$ to $T$, respectively.

[7] When the quality of items associated with fraudulent reviews is 30% lower (i.e., $q^f = 0.7$), their sales are about 14.2%, and 14.3% higher than the sales of the rest of the items in the cases of MF and best-seller recommendations, respectively.

[8] In the presence of MF and best-seller recommendations, items associated with fraudulent reviews have higher sales than items not associated with reviews as long as their quality is higher than 50% of the quality level of nonfraudulent items; if the quality level of items associated with fraudulent reviews is less than or equal to 50%, the sales of nonfraudulent items are higher but there is still a significant transfer of sales from nonfraudulent to fraudulent items compared with cases with no or random recommendations.

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